

# Smart Soil Moisture Monitor

## INTRODUCTION

The Smart Soil Moisture Monitor is an Internet of Things (IoT) based project used to measure the moisture level in soil and monitor it in real time. The system uses a soil moisture sensor to detect whether the soil is dry or wet. The collected data is processed by the Arduino Uno and sent to ThingSpeak cloud through the ESP8266 Wi-Fi module. LEDs are used to indicate soil condition visually. This project helps in smart irrigation and reduces water wastage in agriculture and gardening applications.

## OBJECTIVE

- To monitor soil moisture levels in real time.
- To upload sensor data to the ThingSpeak cloud platform.
- To provide visual alerts using LEDs for dry and wet soil conditions.
- To automate smart irrigation monitoring systems.
- To understand IoT communication using Arduino and ESP8266..

## COMPONENTS REQUIREMENTS

Arduino Uno R3

Soil Moisture Sensor

ESP8266 NodeMCU

Red LED

Breadboard

Jumper Wires

Green LED

USB Cable

Wi-Fi Connection

Android Mobile Phone

## WORKING PRINCIPLE

The soil moisture sensor is inserted into the soil.

The sensor measures the moisture content and produces an analog output.

Arduino Uno reads the sensor value through analog pin A0.

Based on the moisture level:

- Green LED glows when soil is wet.
- Red LED glows when soil is dry.

Arduino communicates with the ESP8266 Wi-Fi module using AT commands.

The ESP8266 sends the soil moisture data to the ThingSpeak cloud platform through Wi-Fi.

# Smart Soil Moisture Monitor

Users can monitor the soil condition remotely using graphs and cloud dashboards.

## SOFTWARE USED

- Arduino IDE
- ThingSpeak
- ESP8266 Board Package

## APPLICATIONS

- Smart irrigation systems
- Agricultural field monitoring
- Home gardening automation

## CODE

```
#include <SoftwareSerial.h>
```

```
SoftwareSerial espSerial(10, 11); // RX, TX
```

```
const int moisturePin = A0;
```

```
const int redLED = 6;
```

```
const int greenLED = 7;
```

```
String apiKey = "3LKQPBANHBWH2LEC";
```

```
String ssid = "V2022";
```

```
String password = "Sheetal2004";
```

```
unsigned long channelID = 3371801;
```

```
void sendATCommand(String cmd, int timeout = 2000) {
```

```
    espSerial.println(cmd);
```

```
    long t = millis();
```

```
    while (millis() - t < timeout) {
```

```
        if (espSerial.available()) {
```

# Smart Soil Moisture Monitor

```
    Serial.write(espSerial.read());
  }
}
}

void setup() {
  Serial.begin(9600);
  espSerial.begin(9600);

  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);

  sendATCommand("AT+RST", 3000);
  sendATCommand("AT+CWMODE=1");
  sendATCommand("AT+CWJAP=\"" + ssid + "\",\"" + password + "\"",
  8000);

  Serial.println("Wi-Fi setup done.");
}

void loop() {

  int raw = analogRead(moisturePin);

  int moisture = map(raw, 1023, 300, 0, 100); // Calibrate for your sensor
  moisture = constrain(moisture, 0, 100);

  Serial.print("Moisture: ");
  Serial.print(moisture);
  Serial.println("%");
```

# Smart Soil Moisture Monitor

```
if (moisture < 30) {  
    digitalWrite(redLED, HIGH);  
    digitalWrite(greenLED, LOW);  
  
    Serial.println("ALERT: Soil too dry!");  
}  
else {  
    digitalWrite(redLED, LOW);  
    digitalWrite(greenLED, HIGH);  
}  
  
// Send to ThingSpeak  
String data = "field1=" + String(moisture);  
  
sendATCommand("AT+CIPSTART=\"TCP\", \"api.thingspeak.com\", 80");  
  
String httpReq =  
    "GET /update?api_key=" + apiKey + "&" + data +  
    " HTTP/1.1\r\nHost: api.thingspeak.com\r\n\r\n";  
  
sendATCommand("AT+CIPSEND=" + String(httpReq.length()));  
  
espSerial.print(httpReq);  
  
delay(1000);  
  
sendATCommand("AT+CIPCLOSE");
```

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```
    delay(20000); // Wait 20 seconds before next reading
  } un();
}
```